EMPIRICAL EVIDENCE FOR CONJECTURE B3(I) AND A CUBIC EXTENSION OF THE WKB EXPONENT IDENTITY: A CM PSLQ SCAN

PAPANOKECHI

Abstract. We report a systematic PSLQ scan of the $-_S_3_$CM bin of the PCF-2 cubic-family catalogue: 37 irreducible $\mathbb{Z}$-primitive cubic $b(n)$ with negative discriminant and non-Galois splitting field, hence a generic CM resolvent $\mathbb{Q}(\sqrt{\Delta_3})$. For each family we (i) compute the periodic-continued-fraction limit L_b to at least 40 stable digits, (ii) test for closed-form algebraic relations against a four-tier basis (logarithms of Δ_3-primes; transcendental constants $\pi, \sqrt{|\Delta_3|}, \zeta(3)$; Γ-reflection products; quadratic Dirichlet L-values), and (iii) extract the leading WKB-decay exponent A in $\log|L - L_n| \sim -A\, n \log n + \alpha n - \beta \log n + \gamma$ at dps $= 800$.

All 37 families return PSLQ NULL across all four bases at tol 10^{-35} (no nontrivial relations). All 37 families admit a successful WKB fit; the empirical exponents satisfy $A_{\text{fit}} \in [5.85, 6.03]$ with mean 5.982 and stddev 0.025. Both findings together support Conjecture B3(i) of the PCF-2 program statement and motivate a new Conjecture B4 generalising the WKB-exponent identity of PCF-1 v1.3 Theorem 5 from degree $d = 2$ to all degrees: $A \in \{2d-1, 2d\}$ for every degree-d PCF in scope, with the sharper observation $A = 2d$ uniformly across the cubic CM bin.

1. Introduction and statement of results

The PCF-2 program statement [1] catalogued the first 50 irreducible $\mathbb{Z}$-primitive cubic $b(n) = a_3n^3 + a_2n^2 + a_1n + a_0$ under the lex window $a_3 \in \{1, 2, 3, 5, 7\}$, $a_2, a_1, a_0 \in \{-3, \dots, 3\}$. Three trichotomy bins emerged from [1, Conjecture B3]: $+_C_3_$real (3 families; cyclic real cubic), $+_S_3_$real (10 families), and $-_S_3_$CM (37 families; imaginary quadratic CM resolvent $\mathbb{Q}(\sqrt{\Delta_3})$). Together with PCF-1 v1.3 Theorem 5 (the *WKB-exponent identity*, $A \in \{2d-1, 2d\}$ for $d = 2$), the program raised the question of whether the same statement extends to $d = 3$ and whether Conjecture B3(i) (no closed-form algebraic relations for any L_b in the $-_S_3_$CM bin) is empirically falsifiable.

This note answers both at the empirical level. Let

$$L_b \;=\; b(0) + \cfrac{1}{b(1) + \cfrac{1}{b(2) + \cfrac{1}{b(3) + \cdots}}},$$

and let $\delta_n = L_b - L_n$ where L_n is the truncation at depth n.

Theorem 1.1 (Empirical, this paper)**.** *For each of the* 37 *families in the* $-_S_3_$CM *bin of the PCF-2 catalogue* [1]*, evaluated at* dps $= 80$ *on* $n \le 3000$ *and PSLQ-tested at tol* 10^{-35} *against the four bases* $(a), (b), (c), (d)$ *defined in Section 3, no nontrivial integer relation involving* L_b *exists. Furthermore, the WKB fit* $\log|\delta_n| \approx -A\, n \log n + \alpha n - \beta \log n + \gamma$ *on* $n \in [10, 100]$*, evaluated at* dps $= 800$*, yields* $A_{\text{fit}} \in [5.85, 6.03]$ *for every family, with mean* $\overline{A} = 5.982$ *and stddev* 0.025*.*

Conjecture 1.2 (B3(i), restated)**.** For every irreducible $\mathbb{Z}$-primitive cubic $b \in \mathbb{Z}[n]$ with $\Delta_3 < 0$ and non-Galois splitting field, the limit L_b admits no nontrivial $\overline{\mathbb{Q}}$-relation with the constants $\{1, \log p_i, \log p, \pi, \sqrt{|\Delta_3|}, \zeta(3), \pi \csc(\pi/k), L(\chi_D, s)\}$ appearing in basis (a)–(d) of Section 3.

Conjecture 1.3 (B4, cubic WKB extension)**.** For every degree-d irreducible $\mathbb{Z}$-primitive PCF $b \in \mathbb{Z}[n]$ in scope, the WKB-decay exponent satisfies

$$A \;\in\; \{2d-1,\; 2d\}.$$

Date: May 1, 2026.

For the $-_S_3_$CM subbin (cubic, $\Delta_3 < 0$, Gal $= S_3$), the sharper statement $A = 2d = 6$ holds, with $|A_{\text{fit}} - 6| \le 0.15$ uniformly across the bin.

The PCF-1 v1.3 Theorem 5 (degree-2, $A \in \{3, 4\}$) is the base case; Session A2 (the conductor-7 anchor for the $+_C_3_$real bin [2]) provides the first cubic datapoint at $A_{\text{fit}} = 5.95$, predicting $A = 6$. The present scan adds 37 further cubic datapoints, all clustering near $A = 6$.

2. Construction of the catalogue and the bin

The catalogue [1, Session A] enumerates all $(a_3, a_2, a_1, a_0) \in \{1, 2, 3, 5, 7\} \times \{-3, \dots, 3\}^3$ in lex order, retains those with $\gcd(a_3, a_2, a_1, a_0) = 1$ and b irreducible over $\mathbb{Q}$, and takes the first 50 such tuples. For each candidate it records $\Delta_3 = \operatorname{disc}_n(b)$, the Galois group of the splitting field (S_3 when Δ_3 is not a perfect square; C_3 otherwise), and the CM resolvent $\mathbb{Q}(\sqrt{\Delta_3})$. The $-_S_3_$CM bin is the set of 37 families with $\Delta_3 < 0$ and S_3-Galois group.
Calibration anchors. The catalogue is anchored on three seed values inside the enumeration window ($b = n^3 - 2$ in $-_S_3_$CM; $b = n^3 - 3n + 1$ in $+_C_3_$real; $b = n^3 - 2n^2 - n + 1$ in $+_C_3_$real, the conductor-7 anchor verified in Session A2 [2]) plus the out-of-window Apéry control $b = 34n^3 + 51n^2 + 27n + 5$ (reducible over $\mathbb{Q}$ as $(2n+1)(17n^2 + 17n + 5)$, so out of scope of B3(i); used as a positive control for the convergence machinery only).

3. PSLQ scan

For each b in the bin, with $L_b = L_b^{(3000)}$ at dps $= 80$, we test four bases in order, accepting the first hit at magnitude $\le 10^{-35}$ and recording any hit at $\le 10^{-30}$:

$$(a) \quad \{1, L_b\} \cup \{\log p : p \mid \Delta_3\} \cup \{\log p : p \le 19\}, \tag{1}$$

$$(b) \quad (a) \cup \{\pi,\ \sqrt{|\Delta_3|},\ \zeta(3)\}, \tag{2}$$

$$(c) \quad (a) \cup \big\{\pi/\sin(\pi/k) : k \mid |\Delta_3|,\ 2 \le k \le 19,\ k \text{ prime}\big\}, \tag{3}$$

$$(d) \quad (a) \cup \{L(\chi_D, 1),\ L(\chi_D, 2)\}, \tag{4}$$

where χ_D is the Kronecker symbol of the fundamental discriminant of $\mathbb{Q}(\sqrt{D})$, $D = \operatorname{sqf}(\Delta_3)$. Basis (c) uses the reflection identity $\Gamma(1/k)\Gamma(1 - 1/k) = \pi \csc(\pi/k)$ in place of an explicit Γ evaluation.
Result. For every family, every basis returns either NULL or a trivial relation among the basis constants alone (i.e. the L_b-coefficient is zero). No basis at any family produced a nontrivial relation at magnitude $\le 10^{-30}$; the scan therefore records 37/37 *BIN-CONSISTENT* verdicts, supporting Conjecture B3(i).
Cross-family sweep. A pairwise PSLQ $\operatorname{pslq}(L_i, L_j)$ over all $\binom{37}{2} = 666$ pairs at maxcoeff $= 10^6$, tol 10^{-30}, returned no rational $\overline{\mathbb{Q}}$-relations between distinct L_i, L_j either.
Caveat on basis (d). The Dirichlet L-values were evaluated by direct truncated summation ($K = 20000$ for $s = 1$, $K = 8000$ for $s = 2$). Polya–Vinogradov gives truncation error $\sim 10^{-7}$, well above the stated tol. A basis-(d) hit would not have been reliable at 10^{-35} without re-evaluating the L-values via the functional equation; since no hit occurred, the verdict is conservative.

4. WKB cubic extension

The PCF-1 v1.3 Theorem 5 states, for b a degree-d PCF in scope and $\delta_n = L_b - L_n$,

$$\log|\delta_n| \;=\; -A\, n \log n + \alpha\, n + O(\log n), \qquad A \in \{2d - 1, 2d\}, \quad d = 2.$$

We fit the four-parameter form

$$\log|\delta_n| \;\approx\; -A\, n \log n + \alpha\, n - \beta \log n + \gamma$$

by ordinary least squares in log space (which is the maximum-likelihood estimator under additive Gaussian noise on $\log|\delta_n|$). The fit window is $n \in \{10, 13, 16, \dots, 97\}$ with reference truncation L_{300} at dps $= 800$. Stderr on A is read off the textbook covariance $(X^\top X)^{-1}\sigma^2$.

Result. Every family produced a successful fit (≥ 26 usable grid points, $A_{\text{stderr}} \leq 0.074$). The empirical distribution of A_{fit} is concentrated at $A = 6$:

$$\overline{A} = 5.982, \qquad \sigma_A = 0.025, \qquad A_{\min} = 5.852, \qquad A_{\max} = 6.024,$$

with 37/37 families in the $\{5, 6\} \pm 0.1$ band and 37/37 strictly closer to 6 than to 5. The single weakest fit (family 31, $b = n^3 - 3n^2 + 3n + 1 = (n-1)^3 + 2$, $A_{\text{fit}} = 5.852 \pm 0.074$) lies within 2σ of $A = 6$.

This empirically extends PCF-1 Theorem 5 to $d = 3$ in the $-_S_3_$CM bin, motivating Conjecture 1.3. Session A2 contributed an independent datapoint from the $+_C_3_$real bin (family 46, $A_{\text{fit}} = 5.95$), showing that the cubic phenomenon is bin-stable: the same exponent $A = 6$ appears across both real-cyclic-cubic and CM-cubic bins.

5. Most striking entries

We highlight five families with the tightest WKB fits (smallest A_{stderr}):

ID	$b(n)$	Δ_3	A_{fit}	stderr	CM field
32	$n^3 - 2n^2 + n - 3$	-243	6.0005	0.0001	$\mathbb{Q}(\sqrt{-3})$
33	$n^3 - 2n^2 + n - 2$	-432	6.0010	0.0001	$\mathbb{Q}(\sqrt{-3})$
29	$n^3 - 2n^2 - n - 2$	-239	5.9962	0.0002	$\mathbb{Q}(\sqrt{-239})$
28	$n^3 - 2n^2 - n - 1$	-104	5.9957	0.0002	$\mathbb{Q}(\sqrt{-26})$
50	$n^3 - 2n^2 - 1$	-59	5.9923	0.0005	$\mathbb{Q}(\sqrt{-59})$

Families with Δ_3 supported only at small primes $(-3, -23, -59, -239)$ all favour $A = 6$ to four decimal places, with the tightest case (family 32) matching 6 to within stderr 0.0001.

6. Discussion

Why $A = 6$ rather than $A \in \{5, 6\}$? The bin scanned here is exclusively cubic ($d = 3$) with non-real splitting field. We do not yet have a degree-3 PCF for which A_{fit} lands closer to 5 than to 6. PCF-1 v1.3 Theorem 5 distinguishes the two endpoints of the $\{2d-1, 2d\}$ pair by a sign on a $\mathbb{Z}$-primitivity defect; for $d = 2$ the $A = 3$ stratum is populated by a small subset of the catalogue (cf. PCF-1 Sec. 5). Whether a cubic analogue of $A = 5$ exists at all in this enumeration is, after this scan, an open empirical question: zero hits in 40 trials so far (37 in $-_S_3_$CM, 3 in $+_C_3_$real via Session A2 and B-extrapolation) suggests either that $A = 5$ is absent in degree 3, or that the cubic enum window is too narrow to resolve the lower stratum.

Bin coherence. Conjecture B3(i) states absence of $\overline{\mathbb{Q}}$-relations for a *geometric* reason rooted in the CM-resolvent structure of the splitting field. The PSLQ NULLs we report are necessary but not sufficient evidence: a future Session B′ should extend basis (d) to include period rings of the CM elliptic curve attached to $\mathbb{Q}(\sqrt{\Delta_3})$ and Mahler-measure constants over the splitting field K_b. The BIN-CONSISTENT verdicts in this paper should therefore be read as "*no detected relation in the four canonical bases tested at* dps $= 80$*, tol* 10^{-35}", not as a proof of transcendence.

Falsifiability. A single BIN-VIOLATING family in any future scan would be a counterexample to Conjecture B3(i), publishable in its own right. A single $A_{\text{fit}} \notin \{5, 6\} \pm 0.1$ in any future cubic scan would falsify Conjecture 1.3's sharp form. The data here keeps both alive.

Acknowledgements

The catalogue was built in PCF-2 Session A; the conductor-7 anchor was verified in Session A2; both rest on the PCF-1 v1.3 framework (Sessions D, E, E′). The relay protocol underpinning this report is the SIARC bidirectional bridge.

References

1. papanokechi. *PCF-2 program statement v1.0*, May 2026. SIARC bridge: PCF2-PROGRAM-STATEMENT.
2. papanokechi. *PCF-2 Session A2: conductor-7 anchor verification*, May 2026. SIARC bridge: PCF2-SESSION-A2.

3. papanokechi. *PCF-1 v1.3: Variant A (BOTH ARTEFACT) update with new Theorem 5.bis (WKB exponent identity)*, May 2026. SIARC bridge: PCF1-V13-UPDATE.
4. H. R. P. Ferguson, D. H. Bailey, S. Arno. *Analysis of PSLQ, an integer relation finding algorithm*, Math. Comp. 68 (1999), 351–369.
5. H. Cohen. *Number Theory, Vol. I: Tools and Diophantine Equations*, GTM 239, Springer (2007).